

# MANIFEST

## BOB-095 — Go-side SSE-broker stress + chaos automation evidence

**Revision:** 1 **Last modified:** 2026-08-19T00:00:00Z

### Scope

qBitTorrent-go/internal/service/sse\_broker\_stress\_chaos\_test.go — §11.4.85 stress + chaos coverage for SSEBroker (internal/service/sse\_broker.go) under go test -race. Go-side sibling to BOB-094's Python tracker-fetch coverage.

### Scenarios delivered (6 tests, all GREEN under -race)

#### STRESS

| # | Test                                                   | Category                                                    |
|---|--------------------------------------------------------|-------------------------------------------------------------|
| 1 | TestSSEBrokerStressChaos_SustainedSubscribeUnsubscribe | sustained-load — 1000 cycles + goroutine-leak guard         |
| 2 | TestSSEBrokerStressChaos_ConcurrentSubscribers         | concurrent — 100 subs × 5 msgs = 500 delivered, no deadlock |
| 3 | TestSSEBrokerStressChaos_BoundaryDoubleUnsubscribe     |                                                             |

| # | Test | Category                                  |
|---|------|-------------------------------------------|
|   |      | boundary —<br>double-close<br>idempotency |

## CHAOS

| # | Test                                                  | Category                                                |
|---|-------------------------------------------------------|---------------------------------------------------------|
| 4 | TestSSEBrokerStressChaos_SlowSubscriberDoesNotBlock   | backpressure<br>isolation —<br>default:<br>clause proof |
| 5 | TestSSEBrokerStressChaos_UnsubscribeDuringPublish     | RLock/Lock<br>ordering<br>under<br>contention           |
| 6 | TestSSEBrokerStressChaos_ProducerFloodBoundedByBuffer | producer-<br>flood<br>bounded by<br>10-msg<br>buffer    |

## §11.4.115 RED-first captures

### Natural RED — §11.4.238 automated-QA-discovers defect

The initial test run of scenario #3 (double-unsubscribe) uncovered a REAL defect the source-side review had missed: `close(ch)` panicked with `close of closed channel`. Captured in `evidence/red_capture_double_unsubscribe.txt`. Fix landed in the same commit: `sse_broker.go` `Subscribe()` gates the `close` on registry membership so a second call is a safe no-op (§11.4.253 idempotency).

### Synthetic RED — backpressure mutation

`sse_broker.go` `Publish()` was TEMPORARILY mutated by removing the `select { ... default: ... }` clause (turning it into a blocking `send`). Scenario #4 wedged as expected; the `-timeout 10s` deadline triggered and the test FAILED. Captured in `evidence/red_capture_backpressure.txt`. The `default: clause` was restored immediately after capture; the GREEN re-run is in `evidence/green_capture_after_restore.txt`.

## Evidence layout

- `latency.json` — p50/p95/p99 + goroutine deltas (scenario 1)
- `concurrent.json` — fan-out totals (scenario 2)
- `boundary_double_unsubscribe.txt` — safe no-op proof (scenario 3)
- `chaos_slow_subscriber.json` — publisher-not-wedged proof (scenario 4)
- `chaos_unsubscribe_during_publish.json` — race + panic counts (scenario 5)
- `chaos_producer_flood.json` — buffer-bounded delivery (scenario 6)
- `red_capture_double_unsubscribe.txt` — §11.4.115 RED (natural, defect-found)
- `red_capture_backpressure.txt` — §11.4.115 RED (mutation, restored)
- `green_capture_after_restore.txt` — final GREEN run, 6/6 PASS

## Discipline citations

- §11.4.85 stress + chaos test types
- §11.4.115(F) machine-written verdict pairs — natural RED / mutation RED
- §11.4.161 rootless — no root/podman needed
- §11.4.14 cleanup on every exit — `defer unsub()` on every `subscribe`
- §11.4.3 SKIP-with-reason — evidence-dir creation failure, never fake pass
- §12/§12.6/§12.12 host safety — `GOMAXPROCS=2 nice -n 19`
- §11.4.238 automated QA discovers — scenario #3 caught a live defect
- §11.4.253 idempotency — the double-unsubscribe fix